Lesson 9.6 - Chebyshev's Theorem

> **Video:** Chebyshev's Theorem by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Statement

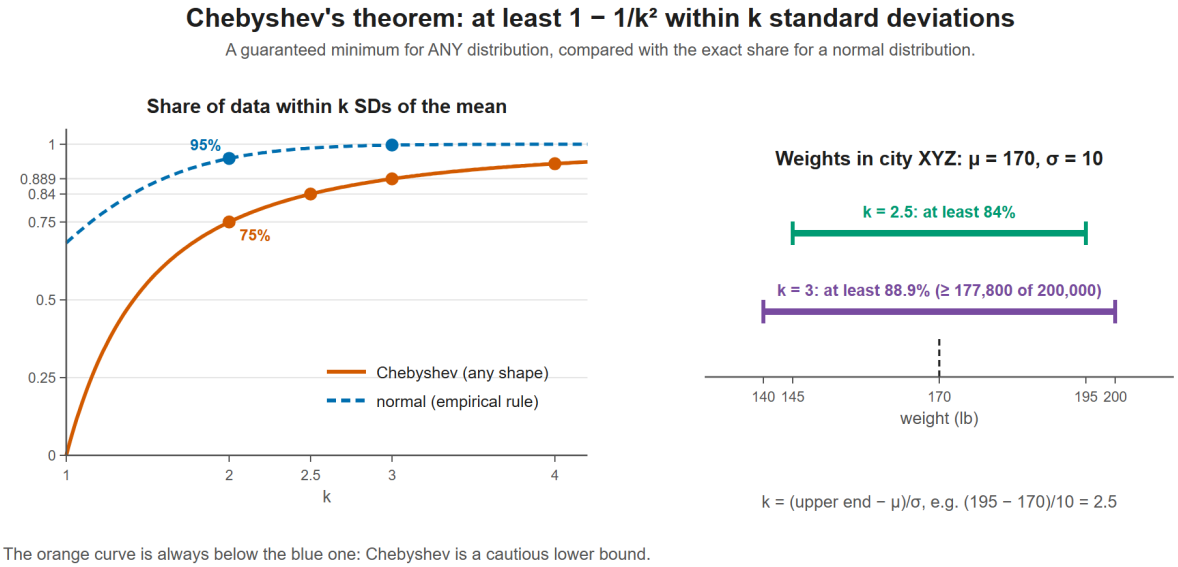
For **any** distribution, whatever its shape, the proportion of data within k standard deviations of the mean is **at least**

$$1 - \frac{1}{k^2}, \qquad k > 1$$

- It gives a **minimum**, not an exact value. The true proportion can be higher.
- It works for **all** distributions, not only normal ones.
- In the video the formula is spoken as "1 over 1 minus k squared". The correct form is $1 - 1/k^2$.

Chebyshev vs the empirical rule

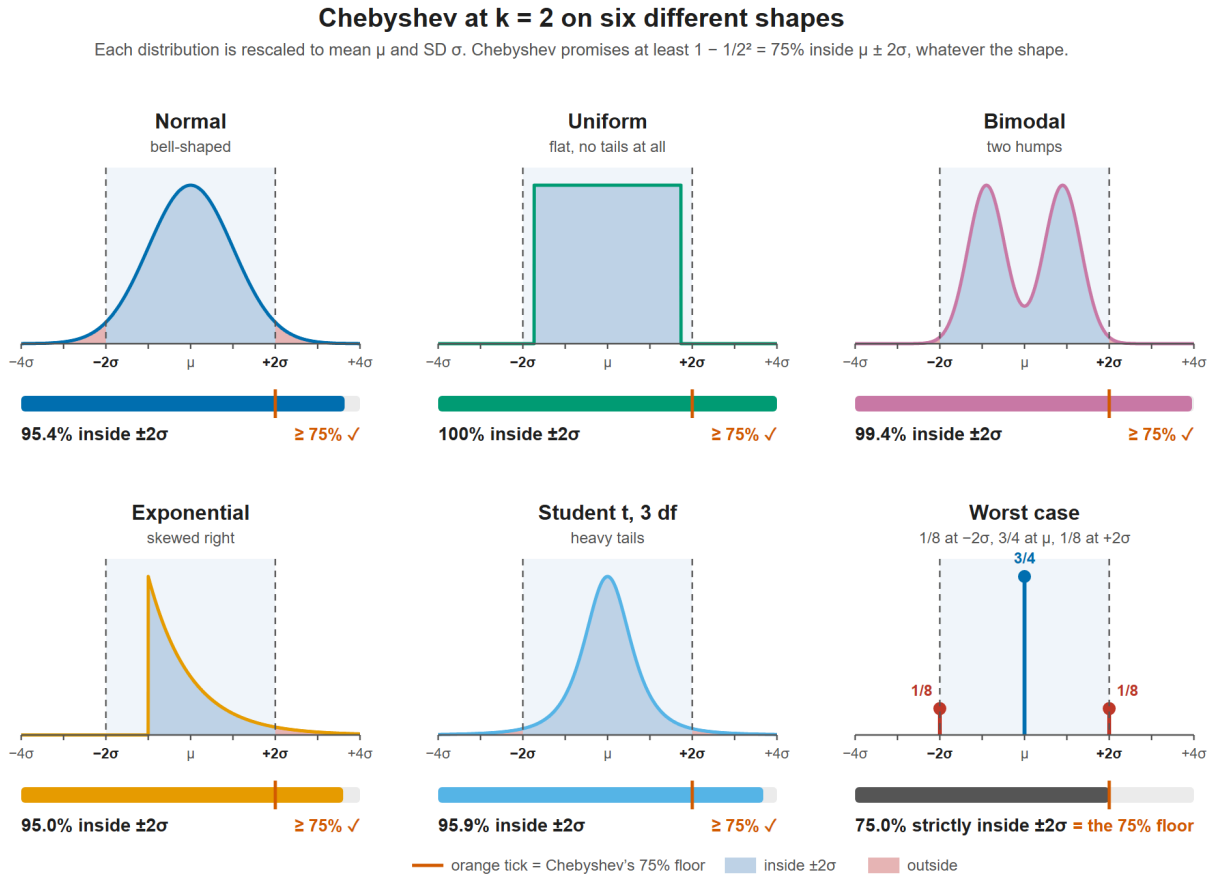
Within	Empirical rule (normal only)	Chebyshev (any distribution, minimum)
1σ	68%	— (k must be > 1)
2σ	95%	$1 - \frac{1}{4} = 75\%$
3σ	99.7%	$1 - \frac{1}{9} \approx 88.9\%$
4σ	≈ 99.99%	$1 - \frac{1}{16} = 93.75\%$



Left: Chebyshev's guaranteed minimum (orange) is always below the exact share for normal data (blue). For example, it guarantees 75% within 2 SDs, while a normal distribution has 95%. Right: the two intervals from the weight example.

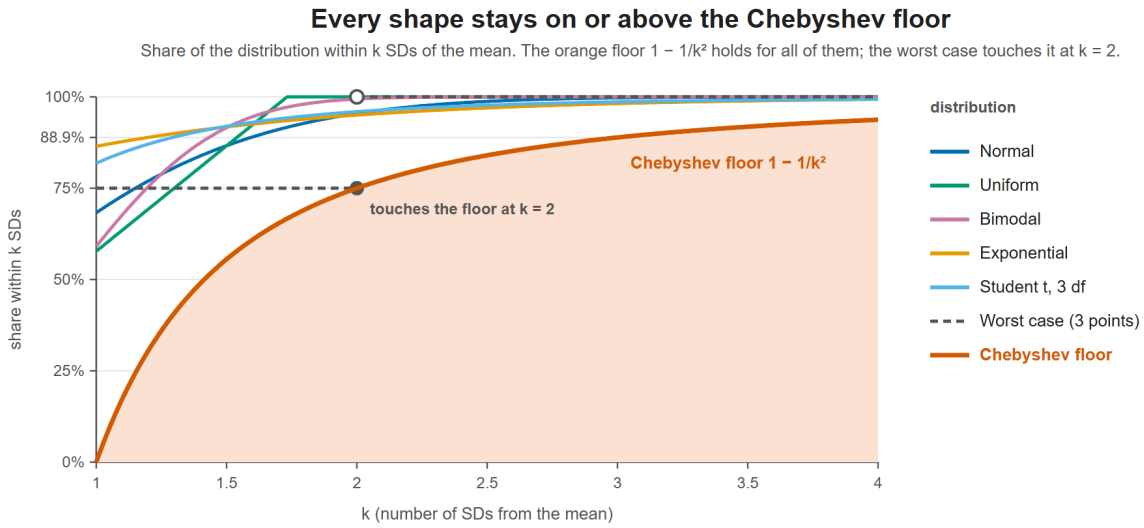
Chebyshev on other famous distributions

"Any distribution" is a strong claim, so here it is tested on six very different shapes. Each one is rescaled to the same mean μ and standard deviation σ, so the axis reads directly in SDs from the mean.



Share within $k\sigma$ of μ	$k = 1.5$	$k = 2$	$k = 3$
Normal	86.6%	95.4%	99.7%
Uniform	86.6%	100%	100%
Bimodal (two humps)	91.6%	99.4%	≈ 100%
Exponential (skewed)	91.8%	95.0%	98.2%
Student t, 3 df (heavy tails)	92.0%	95.9%	98.6%
Worst case (three points)	75%	75%	100%
Chebyshev minimum $1 - 1/k^2$	**55.6%**	**75%**	**88.9%**

- **Every row is at or above the Chebyshev row.** Very different shapes all satisfy the guarantee.
- **Real shapes beat the bound by a lot.** At $k = 2$, five of the six have 95% or more, far above the promised 75%.
- **The bound can't be raised, though.** The worst case puts $\frac{1}{8}$ of the data at $\mu - 2\sigma$, $\frac{3}{4}$ at μ and $\frac{1}{8}$ at $\mu + 2\sigma$. Its SD is exactly σ, since $\sqrt{\frac{1}{8}(2\sigma)^2 + \frac{1}{8}(2\sigma)^2} = \sigma$, yet only 75% lies strictly inside $\mu \pm 2\sigma$. Because one shape lands exactly on 75%, no rule that covers all shapes can promise more.



The orange curve is the floor $1 - 1/k^2$. Each coloured line shows the true share within k SDs for one shape, and none dips below the floor. The dashed worst case touches it at $k = 2$, which is why Chebyshev's theorem is sometimes called "sharp".

What "sharp" means. A bound is *sharp* (or *tight*) when it can't be improved. Suppose someone proposed a stronger rule, such as "at least 80% of any data lies within 2σ". The worst case breaks it, because it has only 75%. So 75% is the best guarantee that holds for every possible shape. Real data usually beats it easily, but no universal rule can promise more.

A worst case for any k. The example above touches the floor only at $k = 2$ because it was built for $k = 2$. The same recipe works for any $k > 1$:

Value	$\mu - k\sigma$	μ	$\mu + k\sigma$
Share of the data	$\frac{1}{2k^2}$	$1 - \frac{1}{k^2}$	$\frac{1}{2k^2}$

- **Its mean is μ**, because the two outer points balance each other.
- **Its SD is exactly σ.** Only the outer points are away from the mean, each by $k\sigma$:

$$\text{variance} = \frac{1}{2k^2}(k\sigma)^2 + \frac{1}{2k^2}(k\sigma)^2 = \frac{\sigma^2}{2} + \frac{\sigma^2}{2} = \sigma^2$$

- **Exactly $1/k^2$ of the data sits $k\sigma$ from the mean**, so the share strictly inside $\mu \pm k\sigma$ is exactly $1 - 1/k^2$, right on the floor.

With $k = 2$ this gives $\frac{1}{8}$, $\frac{3}{4}$, $\frac{1}{8}$, the worst case in the figures. With $k = 3$ it gives $\frac{1}{18}$ at each end and $\frac{8}{9}$ at μ, which touches the floor at 88.9%. So the bound is sharp at every k; each worst-case example just touches the curve at its own k.

Finding k

If an interval is centered on the mean, $\mu \pm k\sigma$, then

$$k = \frac{x - \mu}{\sigma}$$

using the upper end x (the lower end gives the same k).

Example - weights in city XYZ

Mean weight $\mu = 170$ lb, $\sigma = 10$ lb, distribution shape unknown.

(a) Minimum percentage of people weighing between 145 and 195 lb

Both ends are 25 lb from the mean, so

$$k = \frac{x - \mu}{\sigma} = \frac{195 - 170}{10} = 2.5$$

$$1 - \frac{1}{k^2} = 1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 0.84$$

At least 84% of residents weigh between 145 and 195 lb. The real figure could be 86%, 89%, 95%, ...

(b) The city has 200,000 residents. What is the minimum number weighing between 140 and 200 lb?

Both ends are 30 lb from the mean, so

$$k = \frac{x - \mu}{\sigma} = \frac{200 - 170}{10} = 3, \qquad 1 - \frac{1}{k^2} = 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

$$\left(1 - \frac{1}{k^2}\right) \times N = \frac{8}{9} \times 200{,}000 \approx 177{,}778 \text{ people}$$

(The video rounds to 0.889 first and gets 177,800.)

So $P(140 \le X \le 200) \ge 0.889$, and **at least about 177,800** residents weigh between 140 and 200 lb.

Key takeaways

- Chebyshev: at least $1 - 1/k^2$ of any data set lies within k SDs of the mean ($k > 1$).
- Use it when the distribution is **unknown or not normal**. Use the empirical rule only for normal data.
- $k = (x - \mu)/\sigma$ for an interval centered on the mean.
- Answers are **lower bounds**: say "at least".